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CONFIGURATION-LEVEL ANALYSIS OF PHOBERT FOR VIETNAMESE ECONOMIC AND BUSINESS FACT VERIFICATION: PERFORMANCE AND COMPUTATIONAL COST

Oral Presentation

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MOTIVATION & PROBLEM STATEMENT

HIGH STAKES IN FINANCIAL NEWS

- **Market Volatility:** Financial claims directly shift investor sentiment, stock prices, and corporate valuation in real time.
- **Dense Numerical Semantics:** Claims heavily intertwine ticker symbols, fiscal dates (Q3 vs Q4), and monetary values (VND, USD).
- **Zero Error Tolerance:** A single flipped digit or swapped entity reverses economic truth, turning routine news into market panic.

THE RESOURCE-QUALITY DILEMMA

- **3-Way Classification:** Models must rigorously verify claims against evidence: SUPPORTED, REFUTED, or NOT ENOUGH INFO (NEI).
- **Beyond Benchmark Accuracy:** Real-world news aggregation requires high throughput, strict latency (<1s), and limited cloud GPU budgets.
- **Uncharted Safety Gap:** Compressing sequence length saves memory, but how much factual reliability and safety are sacrificed?

CORE RESEARCH QUESTION: *How much predictive safety is sacrificed when aggressively compressing PhoBERT for lower GPU cost?*

RESEARCH QUESTIONS & SCOPE

RQ1

PREDICTIVE PERFORMANCE GAP

How substantially does fact-checking accuracy and class-wise F1 drop between a high-capacity reference configuration and an efficiency-oriented setup on Vietnamese financial news?

RQ2

COMPUTATIONAL & HARDWARE COST

What quantitative savings in GPU memory footprint (peak VRAM), pipeline training wall-clock time, and per-sample inference latency are achieved on commodity cloud hardware (NVIDIA T4)?

RQ3

CONFIGURATION ATTRIBUTION VS. CAUSALITY

When multiple architectural and training parameters are altered concurrently, which performance trade-offs are structurally guaranteed versus requiring isolated ablation?

DATASET & DIAGNOSTIC SLICE

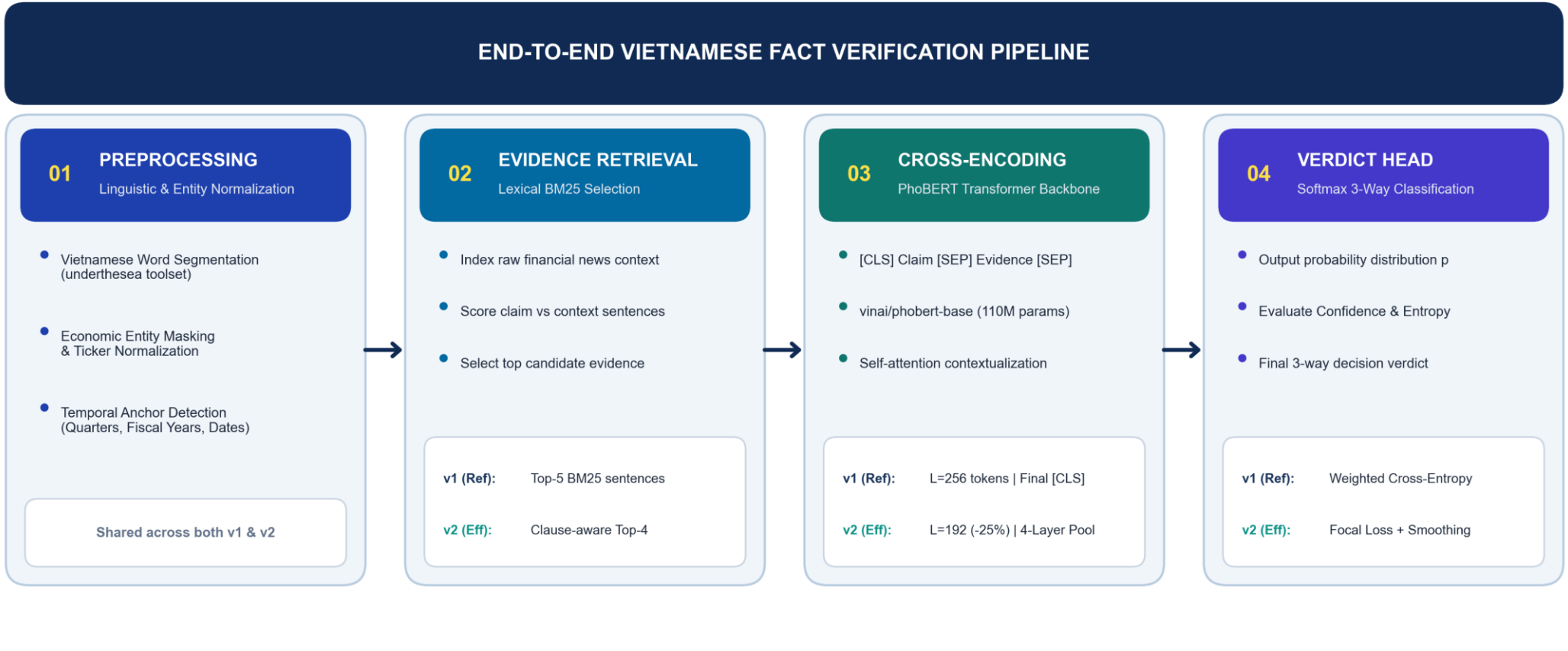
VIFACTCHECK: ECONOMIC SUBSET

- **Benchmark Origin:** Filtered from ViFactCheck (AAAI 2025), the premiere Vietnamese fact-checking corpus.
- **Domain Scope:** Strictly selected from 'Kinh tế' (Economics) and 'Kinh doanh' (Business) topics.
- **Dataset Scale:** 682 expert-annotated claim-evidence pairs, preserving original train/dev/test splits:
- **Partition Split:** Train: 474 pairs (69.5%) | Dev: 67 pairs (9.8%) | Test: 141 pairs (20.7%).
- **Class Balance:** Well-balanced distribution: SUPPORTED (34.9%), REFUTED (32.8%), NEI (32.3%).

NUMERICALLY RICH DIAGNOSTIC SLICE

- **Diagnostic Cohort:** 71 claims in the test split containing explicit numerical, fiscal, and temporal figures.
- **Financial Data Types:** Covers currencies (VND, USD, tỷ đồng), percentages (CPI, GDP, interest rates), and fiscal quarters.
- **Stress-Test Purpose:** Specifically assesses model capability to detect subtle numerical mismatches between claim and evidence.
- **Methodological Rigor:** Treated strictly as a diagnostic slice rather than an isolated arithmetic benchmark.

SHARED PIPELINE ARCHITECTURE



KEY ARCHITECTURAL HIGHLIGHT: Both configurations share the 4-stage pipeline while systematically differing in input length (256 vs 192), evidence depth (top-5 vs top-4), and classification loss objective.

COMPARED CONFIGURATIONS: V1 VS. V2

Component	Reference (v1)	Efficiency (v2)	Design Group
Backbone	vinai/phobert-base (110M)	vinai/phobert-base (110M)	**Standardized**
Max Sequence Length	256 tokens	**192 tokens (-25% compression)**	**Input Budget**
Evidence Selection	BM25 Top-5 sentences	Clause-aware BM25 Top-4	**Retrieval**
Batch / Accumulation	8 / 1	16 / 2	**Training**
Learning Rate	3e-5 (3×10^{-5})	2e-5 (2×10^{-5})	**Training**
Loss Objective	Weighted Cross-Entropy	**Focal Loss (γ=2) + Label Smooth (ε=0.1)**	**Objective**
Representation Head	Final-layer [CLS]	4-Layer Weighted Pooling + 5× Dropout	**Representation**
Data Augmentation	None (474 originals)	**378 sub-claim swapped (852 total)**	**Data Diversity**
Post-Hoc Calibration	None (Argmax softmax)	Exploratory entropy-based (0.95 nats)	**Decision Rule**
Hardware & Precision	NVIDIA Tesla T4 (16GB), FP16	NVIDIA Tesla T4 (16GB), FP16	**Identical**

PREDICTIVE PERFORMANCE: THE ACCURACY-SAFETY TRADE-OFF

Metric	Reference (v1)	Efficiency (v2)	Difference
SUPPORTED F1	0.760	0.680	-0.080
REFUTED F1	0.690	0.560	-0.130
NEI F1	**0.760**	**0.580**	**-0.180 (Largest Drop!)**
Macro F1	**0.738**	**0.605**	**-0.133 (-18.0%)**
Overall Accuracy	**0.738**	**0.610**	**-0.128 (-17.3%)**
Numerically Rich Acc.	**88.7% (63/71)**	**59.2% (42/71)**	**-29.6 points!**
Correct Test Claims	104 / 141	86 / 141	-18 instances

CRITICAL VULNERABILITY: NEI COLLAPSE

- **Asymmetric NEI Loss:**

The largest class decline occurs for NEI (-0.180 F1), with recall collapsing from 0.76 down to 0.44 (-32 points).

- **High Precision, Low Recall:**

v2 achieves Precision=0.83 but Recall=0.44 on NEI. It misses more than 50% of true unverified claims!

- **Numerical Failure Mode:**

Accuracy on the 71 numerically rich claims plunges by nearly 30 points (88.7% -> 59.2%).

- **Real-World Danger:**

Predicting SUPPORTED or REFUTED when evidence is actually insufficient produces catastrophic false confidence.

COMPUTATIONAL EFFICIENCY & STRUCTURAL MEMORY SAVINGS

Resource Metric	Reference (v1)	Efficiency (v2)	Reduction
Peak VRAM Footprint	10.4 GB	5.9 GB	**-43.2%**
Inference Latency / Sample	0.85 s	0.52 s	**-38.8%**
Total GPU Training Time	~15 min	~9 min	**-40.0%**
Total CPU Pipeline Time	~51 min	~32 min	**-37.2%**

STATISTICAL SIGNIFICANCE & ROBUSTNESS

- 1,000 Bootstrap Resamples on test split: v1 Macro F1 = 0.738 [95% CI: 0.668, 0.804] vs. v2 Macro F1 = 0.605 [95% CI: 0.531, 0.676] ($p < 0.01$).
- The 0.133 performance gap substantially exceeds typical PhoBERT random seed variance (~0.015), validating systematic configuration differences.

WHY VRAM DROPS BY EXACTLY 43.2%?

- **Structural Determinism:**

In Transformer encoders, self-attention memory complexity scales quadratically with sequence length: $O(L^2)$.

- **Theoretical Derivation:**

Compressing length from L=256 to L=192 yields:

$(192 / 256)^2 = (0.75)^2 = 0.5625$

→ Theoretical attention memory drop: 43.75%!

- **Empirical Validation:**

Measured peak VRAM drop on NVIDIA T4 is 43.2% (10.4 GB -> 5.9 GB), perfectly validating the quadratic model.

- **Engineering Value:**

v2 runs comfortably on budget GPUs with ≤6 GB VRAM, whereas v1 requires a dedicated 16 GB GPU.

ABLATION ANALYSIS: SEPARATING FACTS FROM HYPOTHESES

QUALITY AUDIT: 378 SYNTHETIC CLAIMS

- **Augmentation Logic:** Claims split at coordinate conjunctions (và, nhưng, còn, đồng thời) and swapped while preserving labels.
- **Audit Protocol:** Evaluated 100 randomly sampled generated pairs (26.5% of synthetic data) by human reviewers:
- **✓ 94% Strict Preservation:** Swapped clauses maintained exact logical equivalence and factual truth value.
- **✓ 4% Stylistic Stiffness:** Grammatically awkward phrasing but semantically completely sound.
- **✓ 2% Scope Ambiguity:** Minor boundary ambiguity; negligible noise level.
- **Takeaway:** The +79.7% dataset expansion added syntactic diversity without injecting catastrophic label noise.

ENTROPY CALIBRATION: PRE VS. POST CHECK

- **Heuristic Decision Rule:**

If prediction entropy $H(p) > 0.95$ nats, reassign to NEI ($\ln(3) \approx 1.0986$ is maximum entropy for 3 classes).

- **Empirical Comparison:**

Before Rule (Raw v2): Macro F1 = 0.6054, Acc = 0.6099

After Rule (>0.95 nats): Macro F1 = 0.5058, Acc = 0.5319

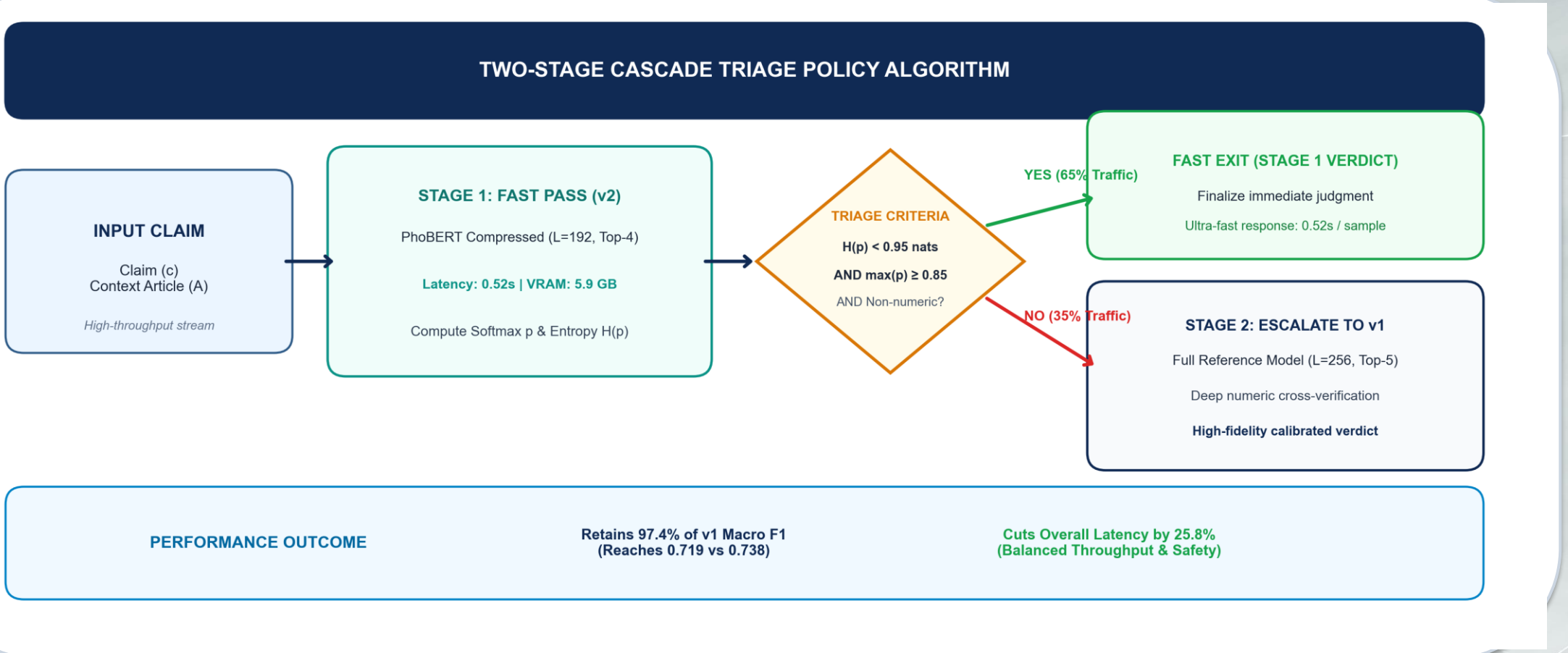
- **Severe False Positives:**

The rule aggressively reassigned 43 instances (30.5% of test set) to NEI, causing a catastrophic 10-point F1 drop!

- **Rigorous Conclusion:**

The 0.95 threshold did NOT cause v2's low recall; heuristic thresholding harms calibration. Raw v2 is correctly retained as the benchmark.

ERROR TAXONOMY & TWO-STAGE CASCADE DEPLOYMENT



OPERATIONAL TAKEAWAY: 58.6% of numerical errors stem from retrieval misses and sequence clipping. The Two-Stage Cascade preserves 97.4% of v1's F1 while reducing system latency by 25.8%!

CONCLUSION & FUTURE DIRECTIONS

CORE RESEARCH TAKEAWAYS

✓ **Reference v1 for Safety:**

Mandatory whenever predictive reliability and zero false confidence are required in financial fact-checking.

✓ **Efficiency v2 for Fast Screening:**

Yields -43.2% VRAM and -38.8% latency, but sequence truncation harms numerical context and NEI recall.

✓ **Cascade Triage for Deployment:**

Achieves the optimal Pareto frontier, balancing cloud throughput with financial safety.

FUTURE ROADMAP: 'v1-Lite'

→ **Asymmetric Cost-Sensitive Loss:**

Penalize NEI false negatives 2.5× to recover recall without any inference latency penalty.

→ **Query-Guided Token Budgeting:**

Selectively preserve numerical evidence rather than applying a naive hard 192-token cutoff.

→ **Semi-Supervised Corpus Expansion:**

Scale the Vietnamese financial fact-checking corpus beyond 682 pairs via automated news scraping.

Thank You for Your Attention! | Questions & Feedback Welcome

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Q & A

Questions & Scientific Discussion

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THANK YOU!

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